OPEN

Comparative Effectiveness of Radiofrequency Ablation, Cryoablation, and Endoscopic Denervation for Lumbar Facet Pain

A Multicenter Trial

Ladislav Kočan, MD, PhD,[a,b] Róbert Rapčan, MD, PhD,[b,c,d] Juraj Mláka, MD,[c] Martin Griger, MD, PhD,[d] Ľubomír Poliak, MD,[b,c] Peter Lenčeš, MD,[b,c] Lenka Kovaličová, BSc,[c] Eva Manik, BSc,[c] Nath Sherdil, MD,[e] Miroslav Burianek, MD,[f] and Janka Vašková, PhD[g]

Study Design. Multicenter, prospective, randomized trial. Trial Registration: ClinicalTrials.gov Identifier: NCT05813639.

Background. Lumbar facet joint syndrome is a common cause of chronic low back pain, often resistant to conservative treatment. Interventional options such as radiofrequency ablation (RF), cryoablation (Cry), and endoscopic facet denervation (ED) are increasingly used, yet head-to-head comparisons are limited.

Objective. To compare the long-term clinical effectiveness of RFA, Cry, and ED in patients with confirmed facet-mediated low back pain.

Methods. A total of 62 patients with chronic lumbar back pain and $\geq 70\%$ relief after controlled diagnostic medial branch blocks were randomized to receive RFA (n = 19), Cry (n = 23), or ED (n = 20). Primary outcomes included changes in back and leg pain intensity (Numeric Rating Scale) and functional disability (Oswestry Disability Index) at 3, 6, 12, and 24 months post-intervention.

Results. All three treatment groups showed significant and sustained improvements in back pain and functional status over the 2-year follow-up. RFA provided the most consistent long-term relief. Cry showed comparable early and mid-term efficacy, though leg pain relief diminished by 24 months. ED achieved rapid early improvement in back pain, but had limited and less durable effects on referred leg pain. No statistically significant differences were observed between groups in any of the outcome measures.

Conclusions. RF, Cry, and ED are all effective and safe treatment modalities for lumbar facet joint syndrome. While RF remains the most established option, Cry and ED may be considered viable alternatives in appropriately selected patients. Treatment should be tailored based on clinical context, patient preferences, and available resources. Further large-scale studies are needed to refine patient selection and optimize outcomes.

Key Words: cryoablation, chronic low back pain, endoscopic denervation, lumbar facet joint syndrome, radiofrequency ablation

(*Spine* 2026;51:1–8)

Chronic low back pain is a prevalent, multifactorial condition with major impact on disability and healthcare costs.[1] Among its sources, the lumbar facet (zygapophysial) joints account for 15% to 45% of cases in various populations.[2] Each facet joint is a synovial articulation forming part of the "three-joint complex" with the intervertebral disc, providing posterior load-bearing and movement control.[3] These joints are richly innervated by the medial branches of the dorsal rami, rendering them vulnerable to pain from degenerative or inflammatory processes.[2,4] Facet joint syndrome most often results from osteoarthritic degeneration, commonly associated with age, mechanical overload, and intervertebral disc disease.[5] Risk factors include repetitive strain, spinal misalignment, facet tropism, trauma, and less commonly, inflammatory

Received for publication July 10, 2025; accepted October 6, 2025.
From the [a]Department of Anaesthesiology and Intensive Medicine, East Slovak Institute of Cardiovascular Disease Faculty of Medicine, Pavol Jozef Šafárik University, Košice, Slovak Republic; [b]EuroPainClinics, Poliklinika Terasa, Košice, Slovak Republic; [c]EuroPainClinics, Bardejov, Bardejov, Slovak Republic; [d]EuroPainClinics, Bratislava – Nové mesto, Slovak Republic; [e]EuroPainClinics, Umeå, Sweden; [f]EuroPainClinics, Prague, Prague 4, Czech Republic; and [g]Department of Medical Biology, Faculty of Medicine, Pavol Jozef Šafárik University, Košice, Slovak Republic, Košice, Slovak Republic.
Signed informed consent was obtained from patients.
The study received approval from the Ethical Committee of the Košice Self-governing Region; Approval No. 6146/2022/ODDZ-25513.
The authors report no conflicts of interest.
Address correspondence and reprint requests to Janka Vašková, PhD, Department of Medical Biology, Faculty of Medicine, Pavol Jozef Šafárik University in Košice, Tr. SNP 1, 040 66 Košice, Slovak Republic; E-mail: janka.vaskova@upjs.sk
Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc. This is an open access article distributed under the terms of the Creative Commons Attribution-Non Commercial-No Derivatives License 4.0 (CCBY-NC-ND), where it is permissible to download and share the work provided it is properly cited. The work cannot be changed in any way or used commercially without permission from the journal.
DOI: 10.1097/BRS.0000000000005539

arthritis.[3] These changes lead to capsular inflammation, cartilage wear, and hypertrophy, activating nociceptive pathways.[1] Clinically, it presents as localized axial back pain without true radiculopathy. Referred pain to the buttocks or posterior thighs may occur but typically does not extend below the knee.[2] Pain is worsened by lumbar extension or prolonged standing and relieved by flexion.[5] While some examination findings (*e.g.* extension-rotation pain) may suggest the diagnosis, they are nonspecific, and imaging often fails to correlate with symptoms.[2]

Given diagnostic uncertainty, controlled medial branch blocks are the reference standard.[1,6] Temporary pain relief after anesthetic injection supports facetogenic pain, though false positives occur in 25% to 45% of cases. Confirmatory blocks and strict criteria ($\geq$80% relief) improve specificity but may miss some true cases,[7] while a $\geq$50% threshold increases sensitivity but risks false positives.[8] Once confirmed, treatment starts conservatively with activity modification, physical therapy, and analgesics; intra-articular steroids may be tried if pain persists, though long-term benefit is limited.[9]

For refractory cases, radiofrequency ablation (RF) of the medial branches is the most established intervention, providing 6 to 12 months of relief in patients with proven facet-mediated pain. A 2022 meta-analysis of 10 RCTs confirmed RF's superiority over sham controls for both short- and long-term relief.[10] Cryoablation (Cry), which induces axonal injury through freezing, has demonstrated short-term efficacy comparable to RF in several retrospective studies,[9] and observational case series.[11] However, regrowth of nerves typically limits its duration of benefit. Though less common than RF, Cry remains a useful option when RF is contraindicated.[9] Recently, endoscopic facet denervation (ED) or "facet debridement" has emerged as an innovative alternative. This technique allows direct visualization and ablation of the medial branch nerves and joint capsule via endoscope.[12] By removing richly innervated structures, it may prolong relief compared with percutaneous RF. A 3-year follow-up study reported $\geq$50% pain reduction in 68% of patients after endoscopic debridement, comparable to RF but potentially more durable.[9]

Lumbar facet joint syndrome is a common contributor to chronic low back pain. Diagnosis relies on anesthetic blocks, and treatment includes conservative, percutaneous, and emerging endoscopic options. This pilot randomized study compares RF ablation, cryoablation, and endoscopic debridement, focusing on pain relief, functional improvement, and long-term outcomes.

MATERIALS AND METHODS

Study Design and Ethics

This study was a multicenter, prospective, randomized, parallel-group, controlled trial designed to compare the effectiveness of three interventional treatments for chronic lumbar facet joint syndrome: radiofrequency neurotomy (RF), cryoablation (Cry), and endoscopic denervation (ED). The trial was conducted in Slovakia across three certified interventional pain management centers under the EuroPainClinics® network, located in Košice, Bardejov, and Bratislava.

The study protocol was approved by the regional ethics committee (Ethical Committee of the Košice Self-governing Region; Approval No. 6146/2022/ODDZ-25513) and registered at ClinicalTrials.gov (ID: NCT05813639). All procedures were conducted in accordance with the Declaration of Helsinki and Good Clinical Practice guidelines. Written informed consent was obtained from all participants before enrollment.

Participants

Patients aged 18 to 80 years with a clinical diagnosis of chronic lumbar facet joint pain were screened for inclusion. Eligibility required chronic low back pain lasting at least 6 months with concurrent leg pain (baseline NRS $\geq$5) and a confirmed diagnosis of facet joint syndrome via two consecutive positive diagnostic medial branch blocks, each producing $\geq$80% pain relief within 12 hours of anesthetic injection.

Exclusion criteria included spinal tumors, spondylitis, vertebral compression fractures, pregnancy or planned pregnancy within 3 years, and any other condition—such as severe comorbidities, noncompliance, or psychological disorders—that the investigators considered likely to interfere with the procedure, follow-up, or reliability of outcome assessment.

Randomization and Masking

A computer-generated randomization list was used to assign eligible patients in a 1:1:1 ratio to the RF, Cry, or ED groups. Randomization was concealed via sealed opaque envelopes provided to the treating physician, who was not involved in outcome assessments. The care provider (interventionist) was not blinded, but the outcome evaluators remained blinded to treatment allocation. Because simple randomization was applied in a relatively small sample (n = 62), minor imbalances between groups occurred. The observed allocation (19, 23, 20) is statistically compatible with chance variation, as with n = 62 and $P = 1/3$ per arm, the expected SD of the per-arm counts is $\approx$3.7. To minimize such imbalances, future trials will employ block or stratified randomization.

Interventions

Patients were enrolled in the study only after a positive diagnostic block performed under fluoroscopic (X-ray) guidance. A block was considered positive if it resulted in $\geq$70% pain relief for the expected duration of the local anesthetic's effect, in accordance with its pharmacokinetic properties. After confirmation of facet-mediated pain, patients were randomized into one of three interventional arms (Table 1, Figure 1).

Radiofrequency Neurotomy (RF group): radiofrequency ablation was performed under fluoroscopic guidance using a monopolar technique. The cannula was advanced in an oblique trajectory toward the junction of the superior articular process (SAP) and transverse process

 Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc.

TABLE 1. Baseline Characteristics of Patients Stratified by Procedure

	RF Group	Cry Group	ED Group
Participants/N	(min-max) med	(min-max) med	(min-max) med
3-months follow-up	19	23	20
6-months follow-up	19	23	20
12-months follow-up	19	23	20
24-months follow-up	19	23	20
Age	(55–75) 68	(54–76) 65	(56–77) 66
Sex (F/M)	(9/10)	(10/13)	(14/6)
ASA	(1–3) 2	(1–3) 2	(1–3) 2
Side of surgery			
Left	4	10	9
Right	5	7	11
Bilateral	10	6	0
Level of surgery			
L1/L2 L2/3, L3/4	1	1	1
L2/3, L3/4, L4/L5	2	2	1
L3/4, L4/L5, L5/S1	15	20	17
L2/3, L3/4, L4/L5, L5/S1	1	0	1

(TP), with position verified in a lateral view. After bone contact, local anesthetic was administered and a radiofrequency probe was inserted. Lesioning was applied at 85°C for 120 seconds per target. Saline injection was used as needed to lower impedance and improve lesion quality. Unipolar lesioning was performed with grounding pads positioned to minimize interference, especially in patients with spinal implants. RF was performed unilaterally or bilaterally at up to four spinal levels, with a total procedure time of approximately 30 minutes.

Cryoablation (Cry group): cryoneurolysis was performed under fluoroscopic guidance using either a perpendicular or near-parallel approach, selected based on patient anatomy. Procedures were carried out under local anesthesia in sterile conditions, with optional mild sedation.

Perpendicular approach: the entry point was identified in a 10 to 15° oblique view at the junction SAP and the transverse process. After local anesthetic infiltration, the cryoprobe was advanced until it contacted bone. Motor stimulation was used selectively to avoid proximity to nerve roots. Near-parallel approach: A caudal tilt of the C-arm (~30°) was applied to align the probe trajectory. The probe was introduced just below the transverse process and advanced cranially along the SAP. Position was confirmed in anteroposterior and lateral views. Motor testing was not routinely performed. In both approaches, two freezing cycles of 120 seconds at –75°C were delivered per target, followed by passive thawing. The procedure was performed unilaterally or bilaterally at up to four spinal levels, with a total procedure time of approximately 30 minutes.

Endoscopic Denervation (ED group): endoscopic facet denervation was performed under local anesthesia (with optional sedation) in sterile operating conditions. Following a <1 cm skin incision, a working cannula was introduced under fluoroscopic guidance to the facet joint level. Using a biportal endoscopic technique, the joint capsule was exposed and incised, and radiofrequency thermal ablation was applied to the lateral aspect of the SAP, targeting terminal nociceptive fibers. In selected cases, denervation extended toward the transverse process to include the medial branch. The procedure was performed unilaterally or bilaterally at up to four spinal levels, with a total procedure time of approximately 30 minutes.

All procedures, including medial branch nerve testing, EF, Cry, and ED were performed under fluoroscopic guidance by physicians certified by the World Institute of Pain (WIP) with the Fellow of Interventional Pain Practice (FIPP) qualification. A total of six FIPP-certified physicians carried out the interventions across the participating centers.

The primary outcomes of the study were change in back pain and leg pain intensity, measured using the Numeric Rating Scale (NRS) ranging from 0 (no pain) to 10 (worst imaginable pain), assessed at 3, 6, 12, and 24 months postprocedure, and change in functional disability, assessed by the Oswestry Disability Index (ODI), ranging from 0 (no disability) to 100 (maximum disability), evaluated at the same time points: 3, 6, 12, and 24 months.

Follow-Up Schedule

Patients were evaluated at baseline and subsequently at 3, 6, 12, and 24 months following the procedure at outpatient pain management clinics. Follow-up data were collected by an independent research team from the Faculty of Medicine, Pavol Jozef Šafárik University in Košice, in collaboration with the Slovak Academy of Sciences in Bratislava.

Statistical Analysis

Descriptive statistics were calculated for baseline characteristics and outcome variables. Normality of distribution was assessed using the Shapiro-Wilk test, and homogeneity of variance using Levene test. Intragroup changes over time were analyzed with paired *t* tests (for normally distributed variables) or Wilcoxon signed-rank tests (for nonparametric data). Between-group comparisons at each time point were conducted using the Kruskal-Wallis test or ANOVA as appropriate. A P-value <0.05 was considered statistically significant. Statistical analysis was performed using SPSS software version 11.0.

The sample size was pragmatically determined based on the anticipated recruitment capacity across the three participating centers during the study period. A formal power-based sample size calculation was not performed, which we acknowledge as a limitation of this trial.

RESULTS

A total of 62 patients with positive diagnostic medial branch block ($\geq 70\%$ relief) met eligibility criteria and were enrolled in the clinical trial. Of these, 19 underwent radiofrequency neurotomy (RF group), 23 received cry-

Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc.

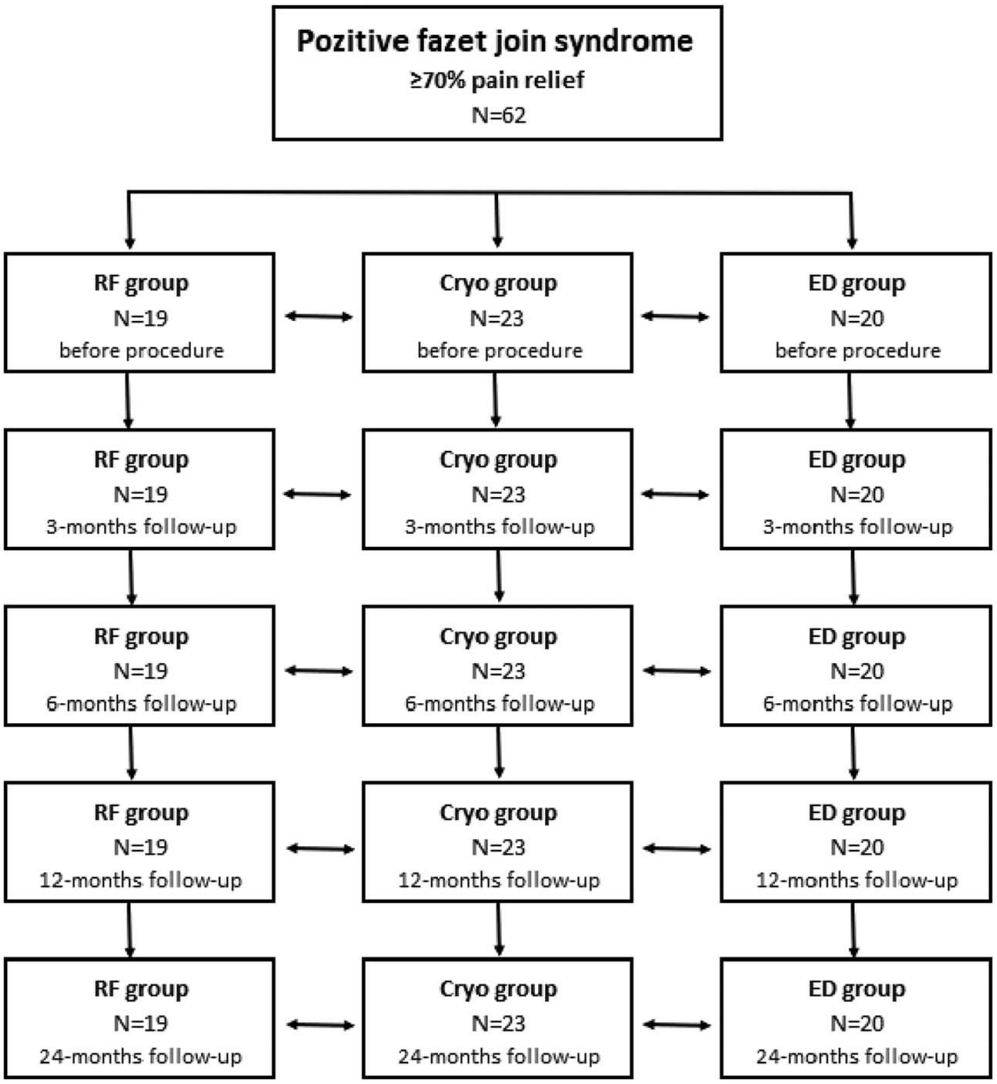

Figure 1. Flow chart of the study patient enrollment.

oablation (Cry group), and 20 were treated with endoscopic denervation (ED group).

Back Pain (NRS)

Back pain intensity, assessed by the Numeric Rating Scale (NRS), significantly decreased in all groups from baseline through 24 months (Table 2). In the RF group, mean back pain decreased from 7.2 ± 1.6 to 4.3 ± 2.6 ($P < 0.001$). The Cry group showed a similar pattern (6.6 ± 1.9 to 5.1 ± 1.9; $P < 0.001$). The ED group achieved the most rapid early improvement (7.2 ± 2.1 to 3.8 ± 2.5 at 3 mo; $P < 0.001$), although partial loss of effect was noted by 24 months (mean 5.7 ± 2.2; $P = 0.011$).

Leg Pain (NRS)

Leg pain improved significantly in the RF group at all follow-up intervals (7.2 ± 2.6 to 4.3 ± 2.6; $P = 0.012$–0.021) (Table 3). In the Cry group, improvement was significant at 3, 6, and 12 months ($P < 0.05$), but not at 24 months (mean 3.8 ± 2.3; $P = 0.088$). In the ED group, improvements in leg pain did not reach statistical significance, suggesting a limited and transient effect on referred leg pain ($P > 0.05$).

Functional Disability (ODI)

All groups demonstrated significant within-group improvement in Oswestry Disability Index (ODI) scores (Table 4). The RF group improved from 63.8% ± 11.6% to 45.7% ± 17.5% at 24 months ($P < 0.001$) within 24 months. The Cry group showed similar benefit (from 59.4% ± 14.5% to 46.1% ± 14.0%; $P < 0.001$). The ED group improved from 58.5% ± 13.4% to 50.5% ± 15.7% ($P = 0.039$), with the largest ODI gain observed at 3 months.

Between-Group Comparison

No statistically significant differences were detected between groups in any primary outcome (back pain, leg

 Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc.

TABLE 2. Back Pain Intensity Expressed by the Numeric Rating Scale (NRS) in Patients Groups With Radiofrequency Neurotomy (RF), Cryoablation (Cry), and Endoscopic Denervation (ED) at 3, 6, 12, and 24 Months Following Procedure

Group (NRS)	Mean (SD)	Median	Min	Max	Comparison	t-statistic	*P*
RF-0 - back pain	7.21 (1.5839)	8	4	10	—		
RF-3 - back pain	4.47 (2.3421)	4	1	9	RF-0 *vs.* RF-3	6.063	0.000
RF-6 - back pain	4.53 (2.5247)	4	1	9	RF-0 *vs.* RF-6	5.064	0.000
RF-12 - back pain	3.74 (2.3533)	3	1	8	RF-0 *vs.* RF-12	6.671	0.000
RF-24 - back pain	4.32 (2.6469)	5	1	8	RF-0 *vs.* RF-24	4.975	0.000
Cry-0 - back pain	6.64 (1.8656)	7	3	10			
Cry-3 - back pain	4.41 (2.2606)	4	1	8	Cry-0 *vs.* Cry-3	5.050	0.000
Cry-6 - back pain	5.09 (1.7157)	4.5	3	8	Cry-0 *vs.* Cry-6	3.986	0.001
Cry-12 - back pain	5.09 (1.8749)	5	2	8	Cry-0 *vs.* Cry-12	4.170	0.000
Cry-24 - back pain	5.14 (1.9832)	5	2	9	Cry-0 *vs.* Cry-24	4.493	0.000
ED-0 - back pain	6.37 (2.1234)	8	2	10			
ED-3 - back pain	4.21 (2.5294)	3	1	9	ED-0 *vs.* ED-3	5.809	0.000
ED-6 - back pain	4.16 (2.3545)	5	1	9	ED-0 *vs.* ED-6	3.682	0.002
ED-12 - back pain	4.42 (2.3866)	5	1	9	ED-0 *vs.* ED-12	4.025	0.001
RF-24 - leg pain	4.74 (2.2321)	6	1	9	ED-0 *vs.* ED-24	2.830	0.011

pain, ODI) at any post-treatment time point (all $P > 0.05$). Although the RF group showed more consistent long-term effects and the ED group demonstrated faster early relief, all three modalities yielded comparable overall outcomes at 2 years (Figure 2).

DISCUSSION

In this randomized trial comparing RF, Cry, and ED for lumbar facet joint syndrome, all groups showed significant pain and disability improvements over 24 months. RF produced the most consistent and durable benefits across pain and ODI measures. Cry offered similar early results, but leg pain relief declined after 12 months, suggesting reduced durability. ED provided rapid back pain relief but did not significantly improve leg pain, indicating limited efficacy for referred symptoms. Despite these differences, no significant between-group differences were found at any time, supporting comparable long-term outcomes and flexibility in treatment choice.

RF neurotomy is well-established for facet-mediated pain, supported by multiple RCTs and reviews. Guidelines classify conventional RF as a level II treatment with moderate recommendation strength.[13] Relief typically lasts 7 to 12 months,[14] with repeat sessions extending benefits. In our cohort, many patients maintained relief for 24 months, likely due to careful selection and effective technique. These results reinforce RF as a reliable first-line option for confirmed lumbar facet joint syndrome.

Like RF, Cry disrupts nociceptive signaling via medial branch ablation, using cold-induced nerve injury. In our study, Cry showed significant improvements similar to RF, with no significant inter-group differences, suggesting comparable mid-term efficacy. Systematic reviews group Cry with RF under facet neurotomy, supported by moderate-quality evidence (level II–III). However, Cry's benefits may be less durable: studies report $\geq 50\%$ pain relief in 72% at 6 weeks, dropping to 57% at 12 months,[15] with only 40% maintaining relief beyond a year.[16] This aligns with the leg pain decline we observed after 12 months, likely from nerve regeneration. The COPE trial similarly found both RF and Cry improved facetogenic pain, with nearly half of patients "much improved" and no serious complications. Though longer-term data are pending, current evidence suggests Cry achieves outcomes close to RF, differing mainly in duration. Cryoablation remains a suitable alternative if RF is contraindicated or declined, though referred leg pain may recur earlier and require sooner reintervention. In our study, both perpendicular and near-parallel cryoprobe techniques had comparable efficacy.[12]

ED (also known as endoscopic rhizotomy or facet arthroscopy) allows direct visualization of the facet joint and medial branch via an endoscope, enabling targeted ablation of nociceptive tissues, osteophytes, and inflamed synovium.[9] Thorough debridement of the joint and capsule, including nerve endplates, may delay reinnervation and provide more durable relief than percutaneous neurotomy.[2] In our study, ED achieved rapid and substantial pain relief, likely from combined nerve ablation and decompression, though leg pain relief declined over time, suggesting partial reinnervation or degenerative progression. These findings align with prior studies: Haufe and Mork reported $\geq 50\%$ pain relief in 68% of patients at 3 years,[9] and Zhu *et al* showed superior outcomes for ED *versus* RF, with median pain-free survival of 20 *versus* 10 months.[17]

Despite promising early outcomes, our two-year results showed no significant advantage of ED over RF or Cry. Given its need for an operating room, sedation, and specialized skills, ED may be best reserved for patients with failed RF or anatomic issues like hypertrophic facets or synovial cysts. Although more invasive, ED has shown low complication rates, with no permanent deficits and minimal wound problems reported.[9]

Our results support ED for select cases but not as a superior treatment for the general facet pain population. Lumbar facet joints, innervated by the medial branches of the dorsal rami, are a common source of chronic low back pain, affecting 15% to 45% of patients.[2] Pain is usually axial and activity-related, sometimes radiating to the

Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc.

TABLE 3. Leg Pain Intensity Expressed by the Numeric Rating Scale (NRS) in Patients Groups With Radiofrequency Neurotomy (RF), Cryoablation (Cry), and Endoscopic Denervation (ED) at 3, 6, 12, and 24 Months Following Procedure

Group (NRS)	Mean (SD)	Median	Min	Max	Comparison	t-statistic	*P*
RF-0 - leg pain	7.21 (2.6079)	7	1	10			
RF-3 - leg pain	4.47 (2.5512)	5	1	9	RF-0 *vs.* RF-3	2.804	0.012
RF-6 - leg pain	4.53 (2.8336)	4	1	9	RF-0 *vs.* RF-6	2.569	0.019
RF-12 - leg pain	3.74 (2.7349)	5	1	9	RF-0 *vs.* RF-12	2.528	0.021
RF-24 - back pain	4.32 (2.6)	5	1	10	RF-0 *vs.* RF-24	2.663	0.016
Cry-0 - leg pain	4.91 (2.8269)	5	1	9			
Cry-3 - leg pain	3 (2.5261)	1.5	1	8	Cry-0 *vs.* Cry-3	3.375	0.003
Cry-6 - leg pain	3.77 (2.4089)	4	1	8	Cry-0 *vs.* Cry-6	2.137	0.044
Cry-12 - leg pain	3.77 (2.7934)	3	1	9	Cry-0 *vs.* Cry-12	2.171	0.042
Cry-24 - leg pain	3.82 (2.3225)	3	1	9	Cry-0 *vs.* Cry-24	1.789	0.088
ED-0 - leg pain	5.06 (2.9255)	6	1	9			
ED-3 - leg pain	3.18 (2.5553)	2	1	9	ED-0 *vs.* ED-3	2.113	0.051
ED-6 - leg pain	4.12 (2.3421)	4	1	9	ED-0 *vs.* ED-6	1.281	0.219
ED-12 - leg pain	4.47 (2.7413)	4	1	9	ED-0 *vs.* ED-12	0.96	0.351
ED-24 - leg pain	4.94 (2.7493)	4	1	9	ED-0 *vs.* ED-24	0.215	0.833

buttocks or thighs.[6] Facet syndrome involves mechanical stress and inflammation, with histologic evidence of cartilage erosion, capsular fibrosis, and inflammatory infiltration.[14] Treatment strategies aim to interrupt nociceptive signaling (RF or Cry), reduce inflammation (intra-articular steroids), or address mechanical pathology (ED). While steroids may offer short-term relief, their long-term efficacy is limited (level IV evidence) compared with neurolytic interventions.[13]

In this study, all patients had chronic, refractory pain confirmed by controlled diagnostic medial branch blocks, improving diagnostic accuracy despite a 25% to 40% false-positive rate. Facet syndrome is difficult to diagnose, as imaging is unreliable and radiographic findings often poorly correlate with symptoms. Controlled blocks remain the gold standard, with significant temporary relief indicating facet-mediated pain. We required a positive block to improve treatment response, though placebo effects or anesthetic spread can still mislead. This underscores the need for strict criteria (*e.g.* >75% relief or dual blocks). Our findings of substantial pain relief across all groups support careful patient selection and confirm that RF, Cry, or ED can provide meaningful, lasting benefit when facet pain is accurately diagnosed.

An important and often underexplored issue is the potential disruption of proprioceptive and mechanoreceptive fibers of the facet joint as a consequence of denervation procedures and capsulotomy. Current anatomic and neurophysiological evidence confirms that facet joints contain mechanoreceptors that provide critical afferent feedback to the central nervous system, contributing to the reflexive activation of deep spinal stabilizers.[18] While facet denervation procedures such as medial branch radiofrequency ablation provide effective analgesia, they may simultaneously disrupt these sensory pathways, potentially leading to transient proprioceptive deficits and even long-term weakening or atrophy of deep paraspinal muscles, such as the multifidus.[19]

Despite the theoretical and clinical importance of this mechanism, it remains poorly addressed in the literature. From a rehabilitative perspective, incorporating physiotherapy aimed at sensorimotor retraining appears warranted. A prospective observational clinical trial demonstrated that adding a 6-week spinal stabilization

TABLE 4. Oswestry Disability Index (ODI) in Patients Groups With Radiofrequency Neurotomy (RF), Cryoablation (Cry), and Endoscopic Denervation (ED) at 3, 6, 12, and 24 Months Following Procedure

Group (ODI)	Mean (SD)	Median	Min	Max	Comparison	t-statistic	*P*
RF-0 - ODI	63.79 (11.6217)	62	42	86			
RF-3 - ODI	48.21 (18.5819)	50	18	82	RF-0 *vs.* RF-3	3.591	0.002
RF-6 - ODI	46.42 (18.3373)	56	18	82	RF-0 *vs.* RF-6	4.314	0.000
RF-12 - ODI	43.58 (18.3857)	38	18	74	RF-0 *vs.* RF-12	4.655	0.000
RF-24 - ODI	45.68 (17.4771)	44	18	76	RF-0 *vs.* RF-24	4.401	0.000
Cry-0 - ODI	59.36 (14.4669)	58	36	92			
Cry-3 - ODI	45.73 (14.0786)	44	20	76	Cry-0 *vs.* Cry-3	5.640	0.000
Cry-6 - ODI	48.64 (15.933)	44	30	88	Cry-0 *vs.* Cry-6	5.632	0.000
Cry-12 - ODI	48.82 (15.6102)	46	30	92	Cry-0 *vs.* Cry-12	4.721	0.000
Cry-24 - ODI	46.09 (14.0269)	42	24	86	Cry-0 *vs.* Cry-24	4.96	0.000
ED-0 - ODI	58.47 (13.4449)	56	36	90			
ED-3 - ODI	44.00 (16.3554)	40	20	78	ED-0 *vs.* ED-3	4.469	0.000
ED-6 - ODI	47.76 (13.9621)	48	20	72	ED-0 *vs.* ED-6	2.997	0.009
ED-12 - ODI	47.53 (16.5458)	48	18	76	ED-0 *vs.* ED-12	2.766	0.014
ED-24 - ODI	50.47 (15.7247)	48	18	78	ED-0 *vs.* ED-24	2.244	0.039

Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc.

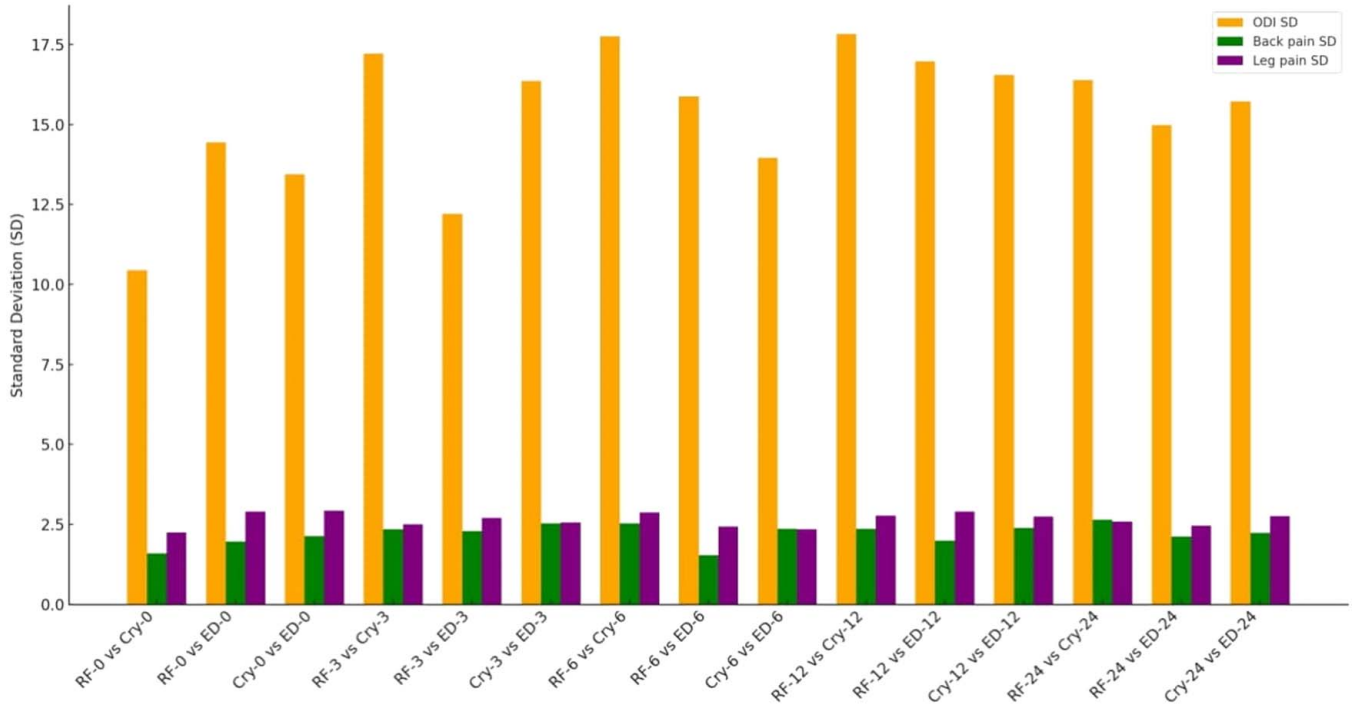

Figure 2. Between-group comparison of pain intensity (VAS) and Oswestry Disability Index (ODI) at 3, 6, 12, and 24 months postintervention.

program following lumbar facet radiofrequency ablation significantly improved pain and functional outcomes.[20] Although our study did not specifically evaluate the impact of rehabilitation, all patients were routinely referred for postprocedural physiotherapy under the supervision of a physiatrist.

In addition, the therapeutic interventions assessed in our study, particularly endoscopic techniques, could theoretically influence spinal segmental stability. Contemporary biomechanical models conceptualize lumbar stability as the integrated outcome of passive structures (vertebrae, intervertebral discs, ligaments, and facet capsules), active musculature, and neural control systems.[21] Disruption of any of these components may result in altered segmental kinematics. Although endoscopic debridement involves partial capsular excision, typically in the inferior quadrant, in our study only unilateral limited decapsulation was performed, which is unlikely to cause clinically significant instability. Moreover, the endoscopic approach is associated with reduced muscle trauma and a lower risk of iatrogenic injury to motor nerve branches compared with open techniques.[22] Future studies should incorporate objective assessments of spinal stability, such as dynamic flexion-extension radiography or kinetic MRI,[23] to detect potential subclinical alterations in segmental biomechanics following these procedures.

This study has several limitations. The relatively small sample size may have limited the ability to detect subtle differences between treatment groups. While within-group improvements were evident, a larger cohort could help determine whether one modality is superior in specific patient subgroups. The open-label design may have introduced bias in patient-reported outcomes. Although the 24-month follow-up period exceeds that of many previous studies, longer-term data are still needed to evaluate the durability of treatment effects.

In addition, no formal power-based sample size calculation was performed; the total enrollment of 62 patients was pragmatically determined by the anticipated recruitment capacity across centers. A modest imbalance in allocation (19:23:20) occurred despite randomization, which is consistent with chance variation in relatively small samples. In future trials, a formal sample size calculation and block or stratified randomization will be applied to ensure adequate power and balanced group allocation.

CONCLUSION

This randomized study shows that RF ablation, cryoablation, and endoscopic facet debridement are all effective for lumbar facet joint syndrome, with no significant differences in long-term outcomes over 2 years. RF remains the most established, while cryoablation offers similar efficacy with some practical advantages. Endoscopic debridement, though more invasive, may help patients with anatomic pathology or failed prior treatments. All three are safe in experienced hands, and treatment should be tailored to patient needs, resources, and goals.

Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc.

➢ Key Points

- ❑ All three interventions significantly reduced back pain and improved functional disability over 24 months, with no statistically significant differences between groups in long-term outcomes, supporting comparable overall effectiveness.
- ❑ Radiofrequency ablation showed the most consistent and durable pain relief for both back and leg pain, confirming its role as a first-line interventional treatment for chronic facet-mediated pain.
- ❑ Cryoablation achieved similar short- and mid-term pain relief compared with RF, but its effects on leg pain tended to decline after 12 months, suggesting reduced durability likely due to nerve regeneration.
- ❑ Endoscopic denervation provided the most rapid early back pain improvement but had limited and transient impact on referred leg pain, making it potentially useful for patients with failed prior RF or with complex facet pathology.
- ❑ Safety and flexibility were demonstrated for all three methods, with no serious adverse events reported, supporting their use in appropriately selected patients based on diagnostic medial branch blocks and individual clinical factors.

References

1. Manchikanti L, Hirsch JA, Falco FJ, et al. Management of lumbar zygapophysial (facet) joint pain. *World J Orthop*. 2016;7:315–337.
2. Perolat R, Kastler A, Nicot B, et al. Facet joint syndrome: from diagnosis to interventional management. *Insights Imaging*. 2018;9:773–789.
3. Kalichman L, Hunter DJ. Lumbar facet joint osteoarthritis: a review. *Semin Arthritis Rheum*. 2007;37:69–80.
4. van Kleef M, Vanelderen P, Cohen SP, et al. Pain originating from the lumbar facet joints. *Pain Pract*. 2010;10:459–469.
5. Curtis L, Dua A, Shah N, et al. Facet Joint Disease, In: Padalia D. ed. *StatPearls*. Treasure Island (FL): StatPearls Publishing; 2025. Available from: https://www.ncbi.nlm.nih.gov/books/NBK541049/
6. Du R, Xu G, Bai X, et al. Facet joint syndrome: pathophysiology, diagnosis, and treatment. *J Pain Res*. 2022;15:3689–3710.
7. Manchikanti L, Pampati S, Cash KA. Making sense of the accuracy of diagnostic lumbar facet joint nerve blocks: an assessment of the implications of 50% relief, 80% relief, single block, or controlled diagnostic blocks. *Pain Physician*. 2010;3:133–143.
8. Schneider BJ, Doan L, Maes MK, et al. Systematic review of the effectiveness of lumbar medial branch thermal radiofrequency neurotomy, stratified for diagnostic methods and procedural technique. *Pain Med*. 2020;21:1122–1141.
9. Haufe SM, Mork AR. Endoscopic facet debridement for the treatment of facet arthritic pain-a novel new technique. *Int J Med Sci*. 2010;7:120–123.
10. Li H, An J, Zhang J, et al. Comparative efficacy of radiofrequency denervation in chronic low back pain: a systematic review and network meta-analysis. *Front Surg*. 2022;9:899538.
11. Kočan L, Rapčan R, Sudzina R, et al. Radiofrequency denervation and cryoablation of the lumbar zygapophysial joints in the treatment of positive lumbar facet joint syndrome - a report of three cases. *Radiol Case Rep*. 2022;17:4515–4520.
12. Yuan HJ, Wang CY, Wang YF. Endoscopic joint capsule and articular process excision to treat lumbar facet joint syndrome: A case report. *World J Clin Cases*. 2021;9:8545–8551.
13. Manchikanti L, Kaye AD, Soin A, et al. Comprehensive evidence-based guidelines for facet joint interventions in the management of chronic spinal pain: American Society of Interventional Pain Physicians (ASIPP) guidelines facet joint interventions 2020 guidelines. *Pain Physician*. 2020;23:S1–S127.
14. Chang CJ, Liu YF, Hsiao YM, et al. Comparison of endoscopic medial branch rhizotomy and radiofrequency ablation in patients with lumbar facet joint pain: a propensity score–matched study. *J Minim Invasive Spine Surg Tech*. 2025;10:S42–S51.
15. Birkenmaier C, Veihelmann A, Trouillier H, et al. Percutaneous cryodenervation of lumbar facet joints: a prospective clinical trial. *Int Orthop*. 2007;31:525–530.
16. Boswell MV, Colson JD, Sehgal N, et al. A systematic review of therapeutic facet joint interventions in chronic spinal pain. *Pain Physician*. 2007;10:229–253.
17. Du T, Lu G, Li J, et al. Pain-free survival after endoscopic rhizotomy versus radiofrequency for lumbar facet joint pain: a real-world comparison study. *Pain Physician*. 2022;25:E87–E94.
18. McLain RF, Pickar JG. Mechanoreceptor endings in human thoracic and lumbar facet joints. *Spine (Phila Pa 1976)*. 1998;23:168–173.
19. Tieppo Francio V, Glicksman M, Leavitt L, et al. Multifidus atrophy and/or dysfunction following lumbar radiofrequency ablation: A systematic review. *PM R*. 2024;16:1384–1394.
20. Cetin H, Kose N, Turkmen C, et al. Do stabilization exercises increase the effects of lumbar facet radiofrequency denervation? *Turk Neurosurg*. 2019;29:576–583.
21. Takahashi M, Iwamoto K, Tomita K, et al. Factors associated with spinal instability in low back lumbar diseases with leg pain: analysis of sagittal translation and segmental angulation. *J Back Musculoskelet Rehabil*. 2023;36:437–444.
22. Rapčan R, Kočan L, Witkovsky V, et al. Endoscopic discectomy of the herniated intervertebral disc and changes in quality-of-life EQ-5D-5L analysis. *Medicine (Baltimore)*. 2023;102:e34188.
23. Michelini G, Corridore A, Torlone S, et al. Dynamic MRI in the evaluation of the spine: state of the art. *Acta Biomed*. 2018;89:89–101.

Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc.